

Course Title:	Calculus
Course Code:	SMTH-101
Semester:	I
Credit Hours:	3 Credit Hours
Pre-requisites:	None

Learning Outcomes

By the end of this course, students will be able to:

1. The students will be able to understand the concept of limit and continuity of the function fundamental concepts of the *derivative* (in terms of tangential slope and instantaneous rate), and *integration*, (in terms of accumulated change and area under the curve)
2. The students will be able to apply various techniques to compute derivatives and integrals of different function types used in optimization problems and other related applications.
3. The students will be able to understand the concept of multi variables functions, partial derivative and double integration.

Course Outline

Unit 1

Introduction to Calculus

Functions, Rule of Four, independent variable, dependent variable, domain, and range of a function, Toolkit Functions: linear, quadratic, polynomial, rational, transcendental functions- Introduction to Limit and continuity: Intuitive and Geometric Approach, techniques of finding limits, Continuous and discontinuous functions, Slope of a function, Average rate of change, Secant Lines

Unit 2

Concept and idea of differentiation

Instantaneous rate of change, Tangent line, Geometrical and Physical meaning of derivatives, Rules of differentiation, Techniques of Differentiation, Product rule, quotient rule, chain rule Applications of derivative:

Increasing decreasing function, Local Maxima and Minima, critical points of a function, First derivative test, Inflection point, Concavity, Second derivative test, Newton Raphson method

Unit 3

Introduction to Integration

The indefinite integral of exponential, polynomials, and trigonometric functions, Techniques of integration (by substitution, by parts), Techniques of integration (by partial fraction), Integration as Area under the curve, Definite Integral, properties of definite integral, Fundamental theorem of calculus, Area between two curves, Trapezoidal Rule, Functions of several variables, Derivatives of several variables, Multiple integrals

- **Teaching-learning Strategies:**
Class Lecture method, which includes seminars, discussions, assignments and projects. (Audio-visual tools are used where necessary)
- **Assignments-Types and Number with calendar:**
According to the choice of respective teacher.
- **Assessment and Examinations:**
According to the University's Semester Rules.

Sr. No.	Elements	Weightage	Details
1	Midterm Assessment	35%	It takes place at the mid-point of the semester.
2	Formative Assessment	25%	It is continuous assessment. It includes: Classroom participation, attendance, assignments, and presentations, homework, attitude and behavior, hands-on-activities, short tests, quizzes etc.
3	Final Assessment	40%	It takes place at the end of the semester. It is mostly in the form of a test, but owing to the nature of the course the teacher may assess their students based on term paper, research proposal development, field work and report writing etc.

Text Books

1. APPLIED CALCULUS by Hughes Hallett (4th Edition)
2. Thomas Calculus by G B Thomas, 11TH Edition.

Suggested Readings

1. CALCULUS by Howard Anton 10th Edition
2. CALCULUS AND ITS APPLICATION by Marvin L. Bittinger 10th Edition
3. Calculus single and Multivariable by Hughes Hallett al., 6th edition.
4. James Stewart, "Multivariable Calculus" 7th Ed.

Course Title:	Introduction to Statistics
Course Code:	STAT-101
Semester:	I
Credit Hours:	3 Credit Hours
Pre-requisites:	N / A

Learning Outcomes

By the end of this course, students will be able to:

1. Understand the key concepts of Statistics.
2. Compute various summary statistics and make use of the graphical techniques to represent data.
3. Know the practical application of Index Numbers along with their computations.
4. Have the basic knowledge of Time Series, its components and the methods to analyze them.

Course Outline

Unit 1

1.1 Introduction

Meaning of Descriptive and Inferential statistics. Population and Sample. Types of variables, Measurement Scales. Sources of Statistical data in Pakistan.

1.2 Tables and Graphs

Description of data by frequency tables and graphs. Stem and Leaf plots and Box plots.

1.3 Measures of Central Tendency

A.M. H.M. G.M., Mode, Median, Quantiles. Properties of Mean with proofs. Weighted Arithmetic Mean. Empirical Relation between Mean, Median and Mode. Relative Merits and Demerits of various averages.

Unit 2

2.1 Measures of Dispersion

Absolute and Relative Measures, Range. Semi-Inter Quartile Range, Mean Deviation, Variance, Standard Deviation. Coefficient of Variation, Coefficient of Mean Deviation, Coefficient of Quartile Deviations, Properties of Variance and Standard Deviation with proofs. Standardized variables, Moments, Moment Ratios, Sheppards Correction, Kurtosis and Skewness.

Unit 3

3.1 Index Numbers

Construction and application of wholesale price Index Numbers. Fixed and chain base methods. Weighted Index Numbers (Laspeyre's, Paasche's Fisher's Ideal and Marshall-Edgeworth Indices). Tests for the consistency of Index Numbers Construction of Consumer price Index Numbers. Sensitive price Indicator.

3.2 Basics of Time series

Components of a time series. Analysis of time series. Measurement of secular trend and seasonal variations by various methods. Deseasonalization of data.

- **Teaching-learning Strategies:**
Class Lecture method, which includes seminars, discussions, assignments and projects. (Audio-visual tools are used where necessary)
- **Assignments-Types and Number with calendar:**
According to the choice of respective teacher.
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Text Books

1. Bluman (2011). *Elementary Statistics* (8th ed.). McGraw Hill, New York.
2. Chaudhry, S.M. & Kamal, S. (2010). *Introduction to Statistical Theory Part I*. Ilmi Kitab Khana, Urdu Bazar, Lahore.

Suggested Readings

1. Beg, M.A., & Mirza, M.D. (2006). *Statistics, Theory and Methods*, Volume I, Carvan Book House, Kutechery Road, Lahore.
2. Clarke G., & Cooke D. (2011). *A basic Course in Statistics* (5th ed.). Arnold Publisher, London.
3. Crawshaw, J., & Chambers, J. (2014). *A concise course in advanced level Statistics with worked examples*, Nelson Thornes, Revised Edition.
4. Johnson, R.A., & Wichern, D.W. (2003). *Business Statistics: Decision making with data*, John Wiley & Sons Inc.
5. Levin, J., & Fox, J.A. (2013). *Elementary Statistics in Social Research* (12th ed.). Pearson Education.
6. Levine, D.M., Kschbiel, T.C., & Berenson, M.L. (2009). *Business Statistics: A first course* (5th ed.). Pearson Education.
7. Macfie, B.P., & Nufrio, P.M. (2006). *Applied Statistics for public policy*, Prentice Hall of India.

GETH-101: ETHICS (FOR NON-MUSLIMS)

General Education Course

Credits:	02
Pre-Requisite:	Nil
Offering:	Undergraduate Degrees (including Associate Degrees)
Placement:	1 – 4 Semesters
Type:	General Education
Fields:	All

DESCRIPTION

This course offers a comprehensive understanding of ethics, various concepts, and the crucial role it plays in different aspects of human life. Students will begin by understanding the fundamental definition of ethics, examining both ancient and contemporary perspectives. The course will also discuss ethical teaching and values of different religions along with concept of Falah.

COURSE LEARNING OUTCOMES

By the end of this course, students will be able to:

1. Define ethics and distinguish between ancient and contemporary ethical concepts.
2. Analyze the importance of ethics in individual, familial, social, economic, and political contexts.
3. Compare and contrast the ethical teachings and values of major world religions.
4. Apply key ethical principles to real-life situations and demonstrate informed decision-making based on these principles.

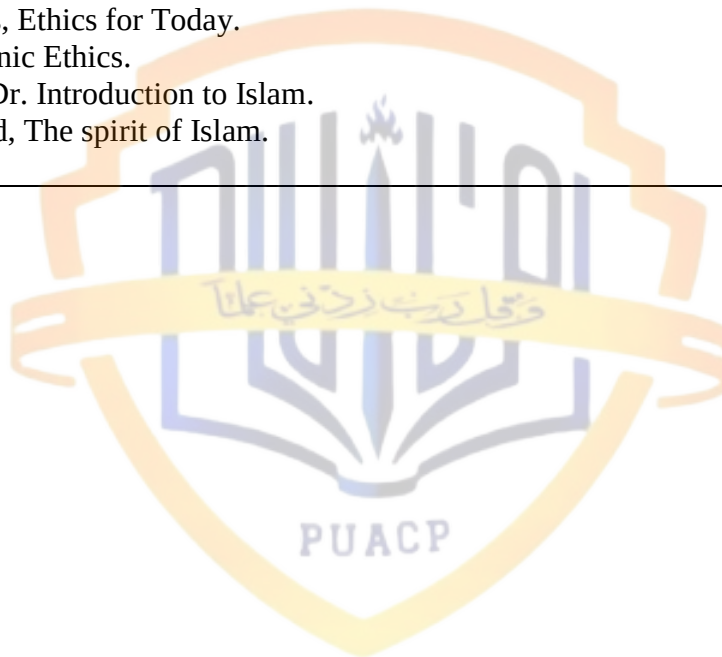
SYLLABUS

1. Definition of Ethics
2. Different concepts of Ethics (Ancient and contemporary)
3. Types of Ethics
 - a. Good Ethics
 - b. Bad Ethics
4. Importance of Ethics in Human Life
 - a. Individual Life
 - b. Family Life
 - c. Social Life
 - d. Importance of Ethics in Economics Life
 - e. Importance of Ethics in Politics
5. Ethical Teachings and Values in Different Religions
 - a. Hinduism
 - b. Buddhism
 - c. Zoroastrianism
 - d. Christianity
 - e. Judaism
 - f. Sikhism
 - g. Islam

6. Ethical Values of the above Religions
 - a. Truthfulness
 - b. Trustworthiness
 - c. Service to Humanity
 - d. Tolerance, Endurance
 - e. Respect for others
 - f. Cooperation, Mutual Help, Selflessness
 - g. Justice (Social Justice, Economic Justice)
 - h. Equality
7. Concept of virtue and Evil in different religions.
8. Concept of “Falah” in Different Religions.
9. Attitude towards other Religions.

RECOMMENDED BOOKS

1. J.S. Mackeuzie, A Manual of Ethics.
2. Harold H. Titus, Ethics for Today.
3. B.A. Dar, Quranic Ethics.
4. Hameedullah, Dr. Introduction to Islam.
5. Ameer Ali Syed, The spirit of Islam.



General Education Course

Credits: 02
Pre-Requisite: Nil
Offering: Undergraduate Degrees (including Associate Degrees)
Placement: 1 – 4 Semesters
Type: General Education
Fields: All

DESCRIPTION

This course is designed to provide students with a comprehensive overview of the fundamentals aspects of Islam, its benefits, practices, history and influence on society. It will further familiarize the students with a solid foundation in understanding Islam from an academic and cultural perspective. Through this course, students will have an enhanced understanding of Islam's multifaceted dimensions which will enable them to navigate complex discussions about Islam's historical and contemporary role, fostering empathy, respect, and informed dialogue.

COURSE LEARNING OUTCOMES

By the end of this course, students will be able to:

1. Demonstrate enhanced knowledge of Islamic foundational beliefs, practices, historical development, fundamental sources of Shiarah, spiritual values and ethical principles.
2. Describe basic sources of Islamic law and their application in daily life.
3. Identify and discuss contemporary issues being faced by the Muslim world including social challenges, gender roles and interfaith interactions.

SYLLABUS

حصہ اول

21 قرآن مجید

- 1- قرآن مجید: معنی و مفہوم، اہمیت و فضیلت، آداب و شرائط تلاوت
- 2- وحی کا معنی و مفہوم، وحی کی اقسام
- 3- قرآن مجید کی جمع و تدوین، حفاظت قرآن
- 2- **حبیبہ**ؓ: نبوی صلی اللہ علیہ وآلہ وسلم
- 1- **حبیبہ**ؓ کا معنی و مفہوم، اقسام **حبیبہ**ؓ (قولی۔ فعلی۔ تقریری)
- 2- ضرورت و اہمیت **حبیبہ**ؓ
- 3- مطالعہ **حبیبہ**ؓ (ترجمہ 6+ تشریح 4=10 نمبر)

1. عَنْ عُمَرَ بْنِ الْخَطَّابِ رَضِيَ اللَّهُ عَنْهُ قَالَ سَمِعْتُ رَسُولَ اللَّهِ ﷺ يَقُولُ: يَهْمُ الْإِسْلَامُ بِالْوَيْلِ، وَيَهْمُ الْوَيْلُ بِمَاتِي، فَمَنْ كَانَتْ هِجْرَتُهُ إِلَى اللَّهِ وَرَسُولِهِ، فَهَجْرَتُهُ إِلَى اللَّهِ وَرَسُولِهِ، وَمَنْ كَانَتْ هِجْرَتُهُ لِيَدِيَا يَصْحَبِي، أَوْ لِعَمَلٍ أَقْبَرُ وَجْهًا، فَهَجْرَتُهُ إِلَى مَا هَاجَرَ إِلَيْهِ (صحيح بخاري: 1)

- [illegible]

16. عَنْ عُمَرَ بْنِ الْخَطَّابِ رَضِيَ اللَّهُ عَنْهُ قَالَ قَالَ رَسُولُ اللَّهِ ﷺ: خَيْرُ نَفْسٍ قَرْنِي نَفْسُ الْيَوْمِ يُؤْمِنُ اللَّهُ الْيَوْمَ (بخاری: 3659)

17. عَنْ جَابِرِ بْنِ عَبْدِ اللَّهِ رَضِيَ اللَّهُ عَنْهُ قَالَ: خَطَبَنَا رَسُولُ اللَّهِ ﷺ وَسَطَرُ أَهْلِ الْعَرَبِ يَخُطِبُهُ الْوَدَاعَ فَقَالَ: يَا أَيُّهَا النَّاسُ إِنَّ رَبَّكُمْ وَاحِدٌ وَإِنِّ آبَاكُمْ لِأَفْضَلُ لِيَعْبُدُنِي عَلَى عَمَلِي لَا لِيَعْبُدُنِي عَلَى عَمَلِي. وَإِنِّي لَأَكْرَهُكُمْ عِنْدَ اللَّهِ أَنْتُمْ أَكْرَهُكُمْ لِي. يَا لَقَدْ وَى لِي. يَا لَقَدْ وَى لِي. يَا لَقَدْ وَى لِي. فَلْيَبْلُغِ اللَّهُ أَهْلَ الْغَايِبِ فَلْيَبْلُغِ اللَّهُ أَهْلَ الْغَايِبِ. (البیهقی، شعب الایمان، باب فی حفظ اللسان، فصل فی حفظ اللسان عن الفخر بأهله)

3- ایمانیات و عبادات: تعارف اور معنی۔

4- سیرت نبوی صلی اللہ علیہ وآلہ وسلم

1- رسول اللہ صلی اللہ علیہ وآلہ وسلم کی حیات طیبہ (حالات زندگی قبل از بعثت، اعلان نبوت، ہجرت مدینہ، غزوات، صلح حدیبیہ، خطبہ جبرہ الوداع)

2- رسول اللہ ﷺ کی مختلف حیثیات: (i) بحیثیت معلم (ii) بحیثیت پیغمبر امن

5- خلافت راشدہ

۱- خلفائے راشدین (حضرت ابو بکر صدیق، حضرت عمر فاروق، حضرت عثمان غنی، حضرت علی المرتضیٰ) کے شخصی احوال

۲- عہد خلافت راشدہ کے اہم خصائص

6- فقہ اسلامی

1- فقہ اسلامی کے مکتبہ: (آن، سنہ، اجماع، قیاس، اجتہاد)

2- ائمہ اربعہ کا تعارف (امام ابو حنیفہ، امام مالک، امام شافعی، امام احمد بن حنبل)

7- اسلام کی سماجی تعلیمات

1- خاندان کا تعارف و اہمیت

2- اسلام میں عورت کا مقام

3- اسلام کا تصور حقوق العباد

4- اسلام کی اخلاقی تعلیمات

8- اسلام اور جدید دنیا

1- عالمگیری - اور اس کے مسائل و تقاضے

2- اسلام و فوج

3- تکثیری معاشرہ Pluralistic Society اور اسلامی تعلیمات

GCCE-101: CIVICS AND COMMUNITY ENGAGEMENT

General Education Course

Credits:	02
Pre-Requisite:	Nil
Offering:	Undergraduate Degrees (including Associate Degrees)
Placement:	1 – 4 Semesters
Type:	General Education
Fields:	All

DESCRIPTION

This course is designed to provide students with fundamental knowledge about civics, citizenship, and community engagement. Students will learn about the essentials of civil society, government, civic responsibilities, inclusivity, and effective ways to participate in shaping the society which will help them apply theoretical knowledge to the real-world situations to make a positive impact on their communities.

COURSE LEARNING OUTCOMES

By the end of this course, students will be able to:

1. Demonstrate fundamental understanding of civics, government, citizenship and civil society.
2. Understand the concept of community and recognize the significance of community engagement for individuals and groups.
3. Recognize the importance of diversity and inclusivity for societal harmony and peaceful coexistence.

SYLLABUS

1. **Introduction to Civics and Citizenship:**
 - Definition of civics, citizenship, and civic engagement.
 - Historical evolution of civic participation.
 - Types of citizenship: active, participatory, digital, etc.
 - The relationship between democracy and citizenship.
2. **Civics and Citizenship**
 - Concepts of civics, citizenship, and civic engagement.
 - Foundations of modern society and citizenship.
 - Types of citizenship: active, participatory, digital, etc.
3. **State, government and Civil Society**
 - Structure and functions of government in Pakistan.
 - The relationship between democracy and civil society.
 - Right to vote and importance of political participation and representation.
4. **Right and Responsibilities**
 - Overview of fundamental rights and liberties of citizens under Constitution of

Pakistan 1973.

- Civic responsibilities and duties.
- Ethical considerations in civic engagement (accountability, non-violence, peaceful dialogue, civility, etc.)

5. **Community Engagement**

- Concept, nature and characteristics of community.
- Community development and social cohesion.
- Approaches to effective community engagement.
- Case studies of successful community driven initiatives.

6. **Advocacy and Activism**

- Public discourse and public opinion.
- Role of advocacy in addressing social issues.
- Social action movements.

7. **Digital Citizenship and Technology**

- The use of digital platforms for civic engagement.
- Cyber ethics and responsible use of social media.
- Digital divides and responsible (access, usage, socioeconomic, geographic, etc.) and their impacts on citizenship.

8. **Diversity, Inclusion and Social Justice:**

- Understanding diversity in society (ethnic, cultural, economic, political etc.).
- Youth, women and minorities' engagement in social development.
- Addressing social inequalities and injustices in Pakistan.
- Promoting inclusive citizenship and equal rights for societal harmony and peaceful coexistence.

SUGGESTED PRACTICAL ACTIVITIES (OPTIONAL)

As part of the overall learning requirements, the course may have one or a combination of the following practical activities:

1. **Community Storytelling:** Students can collect and share stories from community members. This could be done through oral histories, interviews, or multimedia presentations that capture the lived experiences and perspectives of diverse individuals.
2. **Community Event Planning:** Students can organize a community event or workshop that addresses a specific issue or fosters community interaction. This could be a health fair, environmental cleanup, cultural festival, or educational workshop.
3. **Service-Learning:** Students can collaborate with a local nonprofit organization or community group. They can actively contribute by volunteering their time and skills to address a particular community need, such as tutoring, mentoring, or supporting vulnerable populations.
4. **Cultural Exchange Activities:** Students can organize a cultural exchange event that celebrates the diversity within the community. This could include food tastings, performances, and presentations that promote cross-cultural understanding.

SUGGESTED INSTRUCTIONAL / READING MATERIAL

1. “Civics Today: citizenship, Economics, & You” by McGraw-Hill Education.
2. “Citizenship in Diverse Societies” by Will Kymlicka and Wayne Norman.
3. “Engaging Youth in Civic Life” by James Youniss and Peter Levine.
4. “Digital Citizenship in Action: Empowering Students to Engage in Online Communities: by Kristen Mattson.
5. “Globalization and Citizenship: In the Pursuit of a Cosmopolitan Education” by Graham Pike and David Selby.
6. “Community Engagement: Principles, Strategies, and Practices” by Becky J. Feldpausch and Susan M. Omilian.
7. Creating Social Change: A Blueprint for a Better World” by Matthew Clarke and Marie-Monique Steckel.



GENG-101: FUNCTIONAL ENGLISH

General Education Course

Credits:	03
Pre-Requisite:	Nil
Offering:	Undergraduate Degrees (including Associate Degrees)
Placement:	1 – 3 Semesters
Type:	General Education
Fields:	All

DESCRIPTION

This course is designed to equip students with essential language skills for effective communication in diverse real-world scenarios. It focuses on developing proficiency in English language usage: word choices, grammar and sentence structure. In addition, the course will enable students to grasp nuanced messages and tailor their communication effectively through application of comprehension and analytical skills in listening and reading. Moreover, the course encompasses a range of practical communication aspects including professional writing, public speaking, and everyday conversation, ensuring that students are equipped for both academic and professional spheres. An integral part of the course is fostering a deeper understanding of the impact of language on diverse audiences. Students will learn to communicate inclusively and display a strong commitment to cultural awareness in their language use. Additionally, the course will enable them to navigate the globalized world with ease and efficacy, making a positive impact in their functional interactions.

COURSE LEARNING OUTCOMES

By the end of this course, students will be able to:

1. Apply enhanced English communication skills through effective use of word choices, grammar and sentence structure.
2. Comprehend a variety of literary / non-literary written and spoken texts in English.
3. Effectively express information, ideas and opinions in written and spoken English.
4. Recognize inter-cultural variations in the use of English language and to effectively adapt their communication style and content based on diverse cultural and social contexts.

SYLLABUS

1. Foundations of Functional English:

- Vocabulary building (contextual usage, synonyms, antonyms and idiomatic expressions)
- Communicative grammar (subject-verb-agreement, verb tenses, fragments, run-ons, modifiers, articles, word classes, etc.)
- Word formation (affixation, compounding, clipping, back formation, etc.)
- Sentence structure (simple, compound, complex and compound-complex)
- Sound production and pronunciation

2. Comprehension and Analysis:

- Understanding purpose, audience and context

- Contextual interpretation (tones, biases, stereotypes, assumptions, inferences, etc.)
 - Reading strategies (skimming, scanning, SQ4R, critical reading, etc.)
 - Active listening (overcoming listening barriers, focused listening, etc.)
- 3. Effective Communication:**
- Principles of communication (clarity, coherence, conciseness, courteousness, correctness, etc.)
 - Structuring documents (introduction, body, conclusion and formatting)
 - Inclusivity in communication (gender-neutral language, stereotypes, cross-cultural communication, etc.)
 - Public speaking (overcoming stage fright, voice modulation and body language)
 - Presentation skills (organization content, visual aids and engaging the audience)
 - Informal communication (small talk, networking and conversational skills)
 - Professional writing (business e-mails, memos, reports, formal letters, etc.)

PRACTICAL REQUIREMENTS

As part of the overall learning requirements, students will also be exposed to relevant simulations, role-plays and real-life scenarios and will be required to apply skills acquired throughout the course in the form of a final project.

SUGGESTED INSTRUCTIONAL / READING MATERIAL

1. “Understanding and Using English Grammar” by Betty Schramper Azar.
2. “English Grammar in Use” by Raymond Murphy.
3. “The Blue Book of Grammar and Punctuation” by Jane Straus.
4. “English for Specific Purposes: A Learning-Centered Approach” by Tom Hutchinson and Alan Waters.
5. “Cambridge English for Job-hunting” by Colm Downes.
6. “Practical English Usage” by Michael Swan.
7. “Reading Literature and Writing Argument” by Missy James and Alan P. Merickel.
8. “Improving Reading: Strategies, Resources, and Common Core Connections” by Jerry Johns and Susan Lenski.
9. “Comprehension: A Paradigm for Cognition” by Walter Kintsch.
10. “Communication Skills for Business Professionals” by J.P. Verma and Meenakshi Raman.